Business requirements for IoT Smart Agricultural System

Example : Smart Agriculture System

- **1. Connectivity Requirements**
- Use LoRaWAN for long-range communication with sensors in large farm areas.
- Cellular (4G/5G) and Wi-Fi for real-time cloud updates and remote monitoring.
- **2. Data Collection and Processing**
- Soil moisture, temperature, and humidity sensors collect environmental data.
- Data is processed locally on an edge device (Raspberry Pi) for quick decision-making.
- Processed data is sent to the cloud for predictive analytics.

Example : Smart Agriculture System

- **3. Device Management**
- Remote monitoring of sensor status and battery life.
- Over-the-air (OTA) firmware updates for connected devices.
- **4. Security and Privacy**
- Encrypted data transmission (TLS/SSL) to prevent hacking.
- Role-based access control for farm owners and agricultural experts.
- **5. User Interaction**
- Mobile and web dashboard for real-time farm monitoring.
- SMS alerts for critical conditions like drought or excessive humidity.

Example : Smart Agriculture System

- **6. Scalability**
- Ability to add new sensors and devices easily as the farm expands.
- Cloud-based storage ensures unlimited data growth.
- **7. Interoperability**
- Integration with existing irrigation systems using open APIs.
- Compatibility with different sensor brands and communication protocols.
- **8. Reliability and Availability**
- Backup power sources to ensure continuous monitoring.
- Redundant network options (LoRa + Cellular) to prevent connectivity loss.

Example : Smart Agriculture System

- **9. Legal and Regulatory Considerations**
- Compliance with local environmental and agricultural data laws.
- GDPR compliance for user data protection.
- **10. Performance Metrics**
- Response time of the system for sensor data updates.
- Accuracy of predictive analytics in detecting soil dryness.
- **11. Cost Considerations**
- Initial hardware costs (sensors, gateways, cloud storage).
- Operational costs (data transmission, cloud services, and maintenance).